SMART FACTORIES

Twenty-ninth Lecture by Prof. Uday S. Dixit, Department of Mechanical Engineering, IIT Guwahati



Smart Factories – The Future of Manufacturing

- Embracing Industry 4.0
- Basically a cyber-physical system that uses advanced technologies like AI and machine learning to analyze data, automate processes and learn.
- Steps: 1. Data Collection, 2. Data Analysis, 3. Automated Processes, 4. Real-time optimization, 5. Flexibility and adaptability, 6. Integration of technologies

Benefits

- Increased efficiency
- Reduced costs
- Improved product quality
- Enhanced responsiveness
- Better decision-making

What is a Smart Factory?

- Definition: A highly digitized and connected production facility that relies on smart manufacturing.
- Key Traits:
  - Real-time data
  - Autonomous decision-making
  - Adaptive processes

Smart factory technologies

- IoT: connects machines, devices and sensors to collect and share data
- AI
- Machine Learning
- Robotics
- 3D Printing

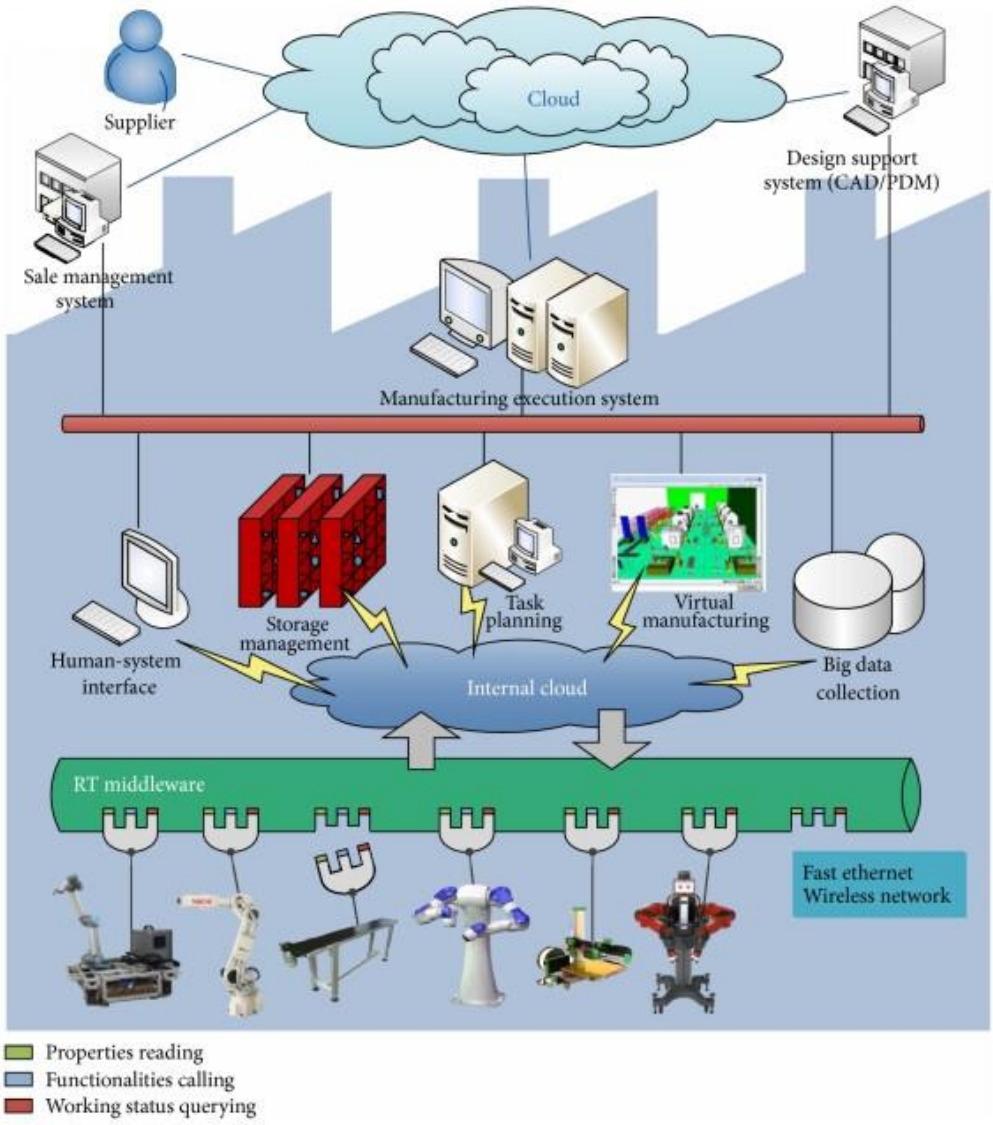
Enabling Technologies

- Sensors
- Cloud Computing
- Big data analytics
- Virtual and Augmented Reality

(Augmented Reality is a digital technology that involves digital information superimposed on reality and viewed via a smartphone. Virtual Reality is a more immersive virtual world that requires special glasses.

Levels of smart factories

- Basic data availability
- Data Analysis
- Active Data
- Action-oriented Data



Wenshan Wang, Xiaoxiao Zhu, Liyu Wang, Qiang Qiu, Qixin Cao., 2016, "Ubiquitous Robotic Technology for Smart Manufacturing System ," Computational Intelligence and Neuroscience
doi:10.1155/2016/6018686

Evolution of Industry

- • Industry 1.0: Mechanization (Water/Steam)
- • Industry 2.0: Mass Production (Electricity)
- • Industry 3.0: Automation (Electronics/IT)
- • Industry 4.0: Cyber-Physical Systems & IoT

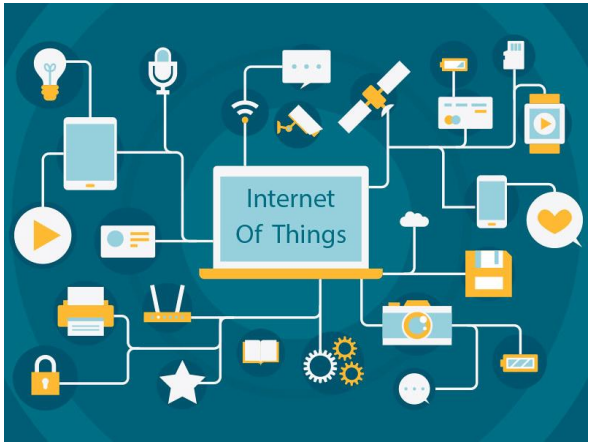
Cyber-Physical Systems

They consist of a collection of computing devices communicating with one another and interacting with the physical world via sensors and actuators in a feedback loop.

Example: Self-driving car

Internet of Things

- IoT describes the network of physical objects with embedded sensors, software and other technologies.
- It is provided a unique identification (UID) and allows data to be transmitted over the network without human interaction objects to collect and exchange data.



https://www.flickr.com/photos/thinkgizmo/36802620122

CPS vs IoT

CPS	IoT
Monitor and control of physical processes in a seamless manner	Network of interconnected devices to collect and exchange data
Integration of physical and cyber components	Network of interconnected devices
More complex	Relatively simple
Sensors and actuators	Embedded sensors
Cyber+Physics	Internet (connectivity) + Things

Key Enabling Technologies of Smart Factories

- Internet of Things (IoT)
- Cyber-Physical Systems (CPS)
- Artificial Intelligence (AI) & Machine Learning (ML)
- Cloud & Edge Computing
- Big Data Analytics
- Robotics & Automation
- Digital Twin
- 5G Connectivity

Edge Computing

- It is a distributed computing framework that brings data processing and storage closer to the source of the data, often at the network edge, where users and devices access it.
- It reduces latency (delay and time lag between request and response).

Smart Factory Architecture

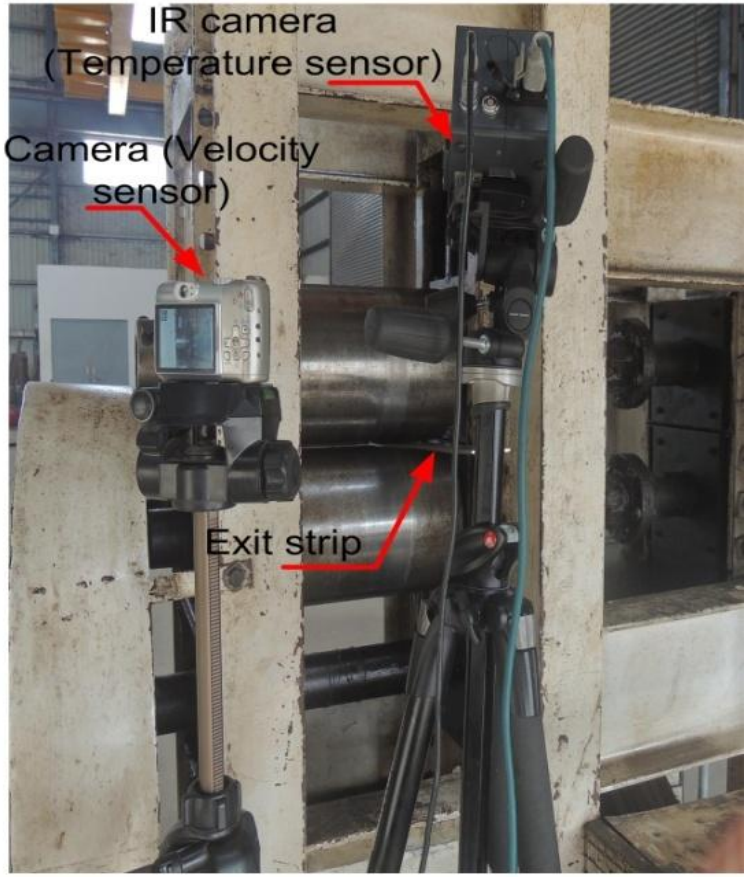
- • Layers:
- - Physical Layer: Sensors, actuators
- - Network Layer: Connectivity (5G, Wi-Fi, LoRa)
- - Data Layer: Storage, databases
- - Application Layer: Dashboards, analytics
- • Cybersecurity as a cross-layer concern

Real-Time Monitoring with IoT

- Use of sensors and devices to collect machine data
- Example: Temperature, vibration, energy usage
- Data helps in predictive maintenance and efficiency



(a)



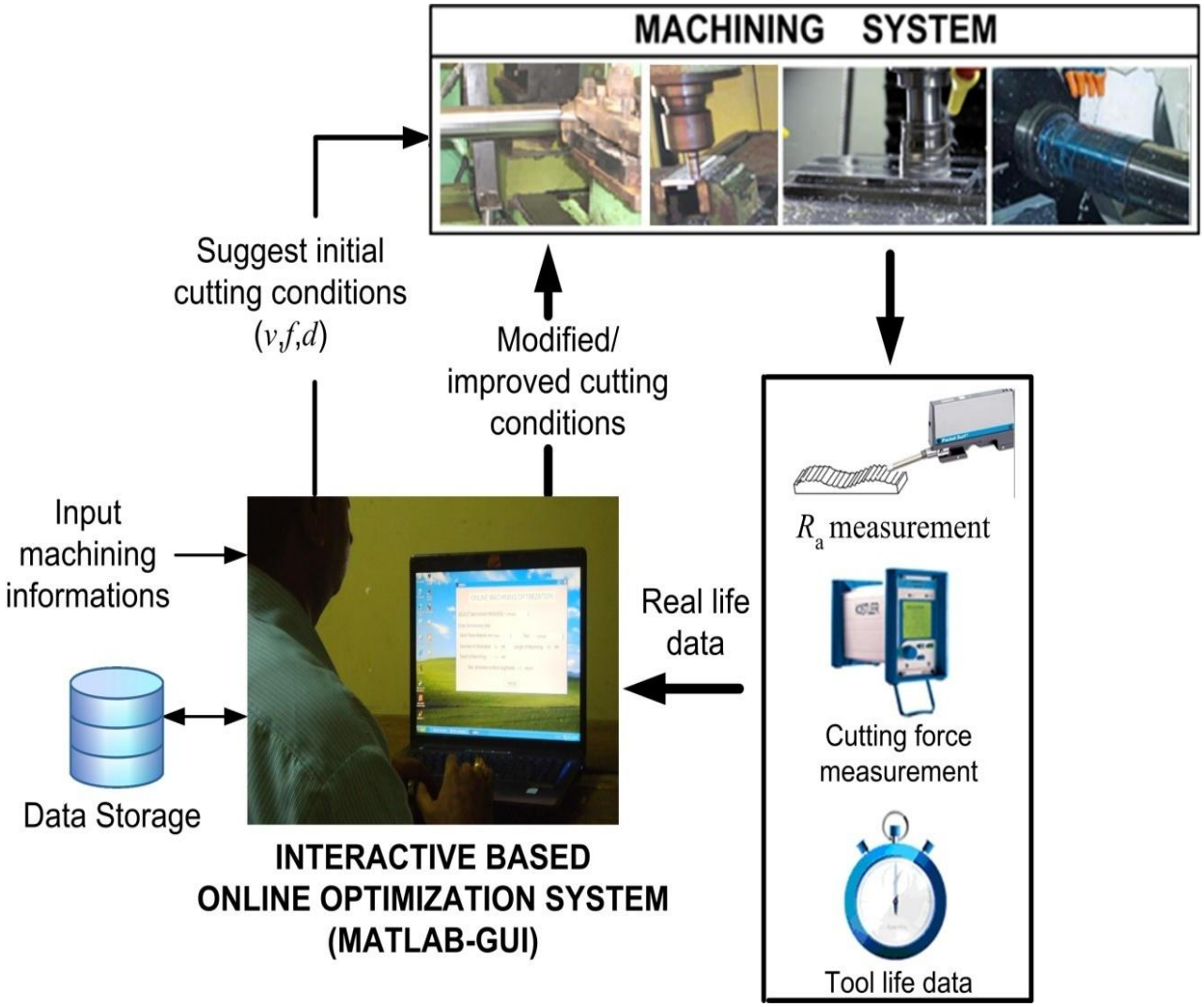
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Digital Twin Concept

- • Virtual representation of a physical process/system
- • Enables simulation, monitoring, and optimization
- • Example: Simulating a factory line before actual production

AI & Machine Learning Applications

• Predictive maintenance • Quality inspection via computer vision • Process optimization • Demand forecasting



Case Study 1

- **Bosch Smart Factory (Germany & India)**
- **What they do:** Use IoT sensors, AI, and predictive maintenance.
- **Key feature:** Reduced downtime by 25%, improved quality control.
- **Location:** Stuttgart, Germany; Nashik, India.
- **Source:** Bosch press releases, Bosch Industry 4.0

2. Siemens Amberg Plant (Germany)

- **What they do:** Produces Siemens automation products.
- **Key feature:** Over 75% of the value chain is automated, with a defect rate of just 12 per million.
- **Highlight:** Considered a global model for Industry 4.0.
- **Source:** Siemens Newsroom and official Siemens Industry 4.0 case studies.

3. GE Brilliant Factory (USA, Pune in India)

- **What they do:** Combine advanced analytics with lean manufacturing.
- **Key feature:** Use of digital twins, real-time data, and AI for process optimization.
- **Source:** General Electric's website, Harvard Business Review.

4. Tata Steel Kalinganagar, Jajpur, Odisha (India)

- **What they do:** Implemented smart sensors, AI/ML for quality and maintenance.
- **Key feature:** Real-time control over blast furnace operations, improving energy efficiency.
- **Source:** Tata Steel sustainability reports and industry news.

5. Schneider Electric Smart Factory (Hyderabad, India)

- **What they do:** Use IIoT-enabled EcoStruxure™ platform.
- **Key feature:** Real-time energy monitoring and predictive maintenance.
- **Recognition:** Part of the World Economic Forum's Global Lighthouse Network.
- **Source:** Schneider Electric site and WEF reports.

Benefits of Smart Factories

- • Improved efficiency and uptime
- • Enhanced product quality
- • Reduced operational costs
- • Better energy management
- • Greater flexibility and customization

Challenges in Implementation

- • High initial cost
- • Cybersecurity vulnerabilities
- • Workforce skill gap
- • Integration with legacy systems
- • Data ownership and privacy

Future Trends

- Integration with quantum computing
- Human-robot collaboration
- Sustainable and green smart factories

Conclusion

- • Smart factories are intelligent, adaptive, and data-driven
- • They transform traditional manufacturing
- • Need a roadmap for adoption: technology, training, and transformation